Package 'rENM.model'

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Type Package

Title Modeling tools for the rENM framework

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Description The rENM framework is a modular sysem for reconstucting and analyzing long-term ecological niche dynamics using time-indexed environmental data and species occurrence records. Its packages support climate-based niche modeling, temporal analysis of climatic suitability, and reproducible workflows for studying ecological change.

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Imports rENM.core,
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foreach,
ggplot2,
maxnet,
raster,
scales,
sdm,
sf,
sp,
terra,
usdm

Contents

`create_ensemble_model` *Build an ensemble ENM and derived products for a species*

Description

Constructs an ensemble ecological niche model (ENM) for a given species (four-letter banding code) and a 5-year time slice using the sdm framework, producing ensemble predictions and derived products including range maps and model diagnostics.

Usage

```
create_ensemble_model(
  alpha_code,
  year,
  methods = c("maxnet", "rf", "brt", "glm", "mars"),
  reps = 3,
  bg = 2500,
  tp = 40,
  op = 2,
  ensemble_method = "unweighted",
  io = c("raster", "terra"),
  overwrite = TRUE,
  verbose = TRUE
)
```

Arguments

`alpha_code`	Character. Four-letter banding code, case-insensitive (validated as exactly four A-Z letters; stored upper-case).
`year`	Integer. A 5-year bin start in (1980, 2020) divisible by 5.
`methods`	Character. SDM algorithms to fit. Defaults to `c("maxnet","rf","brt","glm","mars")`.
`reps`	Integer. Number of replicated subsampling runs. Default `3`.
`bg`	Integer. Number of background points. Default `2500`.
`tp`	Numeric. Test partition percent (0-100). Default `40`.
`op`	Integer. Threshold optimization option passed to `sdm::getEvaluation` for binarization; 2 = max(se+sp). Default `2`.

ensemble_method — Character. Ensemble combination method passed to `sdm::ensemble` in `setting = list(method = ...)`. Default `"unweighted"`.

io — Character. Raster I/O backend for writing outputs; one of `"raster"` or `"terra"`. Predictors are read via raster either way to maintain sdm compatibility. Default `"raster"`.

overwrite — Logical. Overwrite existing output files. Default `TRUE`.

verbose — Logical. Emit timestamped console progress. Default `TRUE`.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context: Inputs/paths must follow the rENM run structure:

```
<rENM_project_dir()>/runs/<ALPHA>/TimeSeries/<YEAR>/occs/of-<YEAR>.csv
<rENM_project_dir()>/runs/<ALPHA>/TimeSeries/<YEAR>/vars/*.asc
<rENM_project_dir()>/runs/<ALPHA>/TimeSeries/<YEAR>/model/  (outputs written here)
```

Inputs: The occurrence CSV must include `longitude`, `latitude`.

Methods: Models are trained via `sdm::sdm()` with replicated subsampling (`reps`) and a test partition (`tp`, percent). Background points (bg) are drawn using `method = "gRandom"`.

The default modeling ensemble uses `"maxnet"` (a native R implementation of maximum entropy modeling) instead of `"maxent"` to avoid Java dependencies and improve reproducibility in CRAN-compliant environments.

Outputs:

- Ensemble prediction rasters (GeoTIFF and ASCII)
- Model statistics and variable importance summaries
- Binary range map derived via threshold optimization

Logging: A human-readable, append-only summary is written to: `<rENM_project_dir()>/runs/<alpha_code>/_log.txt` with key parameters, file outputs, and per-year status. Console messages include timestamps for traceability.

Value

Invisibly returns a List with structured elements:

- `data`: sdmData object containing training data
- `model`: sdmModel object representing fitted models
- `ensemble`: RasterLayer of predicted suitability
- `range`: RasterLayer of binary presence/absence
- `paths`: List of important written file paths
- `stats`: List of captured model statistics and variable importance outputs

Side effects include writing raster outputs, statistics files, variable importance summaries, plots, and appending a log entry to the run log file.

See Also

sdm, sdmData, ensemble, getEvaluation

Examples

```
## Not run:
# Default knobs; raster I/O
result <- create_ensemble_model("CASP", 2000)

# Heavier replication, custom methods, terra I/O
result <- create_ensemble_model(
  alpha_code = "CASP", year = 2000,
  methods = c("maxnet","glm","brt","rf","mars"),
  reps = 5, bg = 5000, tp = 30, op = 2,
  ensemble_method = "unweighted",
  io = "terra", overwrite = TRUE, verbose = TRUE
)

## End(Not run)
```

---

create_range_map — *Create and save a binary presence-absence range map for a species*

Description

Build a ggplot2 map for a binary (0/1) presence-absence raster and save the plot image and raster outputs in multiple formats. Supports multiple raster input types and integrates species range overlays.

Usage

```
create_range_map(
  pa,
  alpha_code,
  year,
  state_shp = file.path(rENM_project_dir(),
    "data/shapefiles/tl_2012_us_state/tl_2012_us_state.shp"),
  title = "Range Map",
  layer = 1L,
  minor_div = 2L,
  width_px = 1000,
  height_px = 750,
  dpi = 72,
  title_size = 25,
  legend_size = 15,
  legend_title_size = 15,
  axis_title_size = 20,
  axis_text_size = 18
)
```

Arguments

`pa` SpatRaster, Character, or Raster. A presence-absence raster.

`alpha_code` Character. Code used to construct output paths and to look up GAP.RANGE via get_species_info(alpha_code) for locating the species range shapefile at: `<rENM_project_dir()>/data/shapefiles/<gap_range>/ <gap_range>.shp`. Must be a non-empty string (e.g., "casp").

`year` Integer, Character. Year inserted into the output path and name.

`state_shp` File path. Path to a states or administrative boundaries shapefile. Defaults to: `<rENM_project_dir()>/data/shapefiles/tl_2012_us_state/ tl_2012_us_state.shp`.

`title` Character. Plot title. Default "Range Map".

`layer` Integer. Layer index to use when pa has multiple layers. Default 1L.

`minor_div` Integer. Number of minor grid divisions between major axis breaks. For example, 2L places two minor ticks between majors. Default 2L.

`width_px`, `height_px`
Numeric. Output PNG dimensions in pixels. Defaults 1000 and 750.

`dpi` Numeric. Plot DPI used to convert pixels to inches for ggsave(). Default 72.

`title_size`, `legend_size`, `legend_title_size`, `axis_title_size`, `axis_text_size`
Numeric. Font sizes for plot elements.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context Constructs a presence-absence map and raster exports within the run-specific directory: `<rENM_project_dir()>/runs/<alpha_code>/TimeSeries/<year>/model/`

Inputs Presence (1) is rendered in orange; Absence (0) in light gray. NA cells are dropped from the plot. If pa has multiple layers, layer selects which to use.

Species range source (package helper) This function does not search for range shapefiles dynamically. It calls the internal helper `get_species_info(alpha_code)`, looks up GAP.RANGE in `<rENM_project_dir()>/data/_species.csv`, and loads the shapefile at: `<rENM_project_dir()>/data/shapefiles` `<gap_range>.shp`

Methods Values in the raster are coerced to binary by rounding (round()) and then mapped to the factor levels "Presence" (1) and "Absence" (0) for plotting. The vector layers (states and species range) are reprojected to the raster CRS if necessary. Axis breaks are computed with pretty() plus configurable minor divisions.

Outputs The following files are written:

- Plot image: <fn>.png
- Raster (AAIGrid): <fn>.asc
- Raster (GeoTIFF): <fn>.tif

After writing .asc and .tif, common sidecar files (.prj, .aux.xml) are removed if present to keep the directory clean.

Value

Invisibly returns a List with:

- plot: ggplot2 object representing the rendered map
- img_path: Character path to the saved .png file
- asc_path: Character path to the saved .asc file
- tif_path: Character path to the saved .tif file

Side effects include writing raster and image files to disk and removing associated sidecar files when present.

See Also

[terra::rast()], [terra::writeRaster()], [sf::st_as_sf()], [ggplot2::ggplot()]

Examples

```
## Not run:
  # Using a file path:
  create_range_map(
    pa         = "rasters/casp_1980_pa.tif",
    alpha_code = "casp",
    year       = 1980
  )

  # Using a SpatRaster and a specific layer:
  r <- terra::rast("rasters/casp_stack.tif")
  create_range_map(r, "casp", 1990, layer = 2L,
                   title = "CASP 1990 Range")

## End(Not run)
```

`create_timeseries` *Build an ensemble rENM time series for a species*

Description

Schedules and executes create_ensemble_model() for each 5-year time bin between 1980 and 2020 in parallel, producing a full rENM time series for a given species.

Usage

```
create_timeseries(alpha_code, project_dir = NULL)
```

Arguments

`alpha_code`	Character. Four-letter banding code, e.g., "CASP".
`project_dir`	Character, NULL. Base directory for outputs and logs. If NULL, the directory is resolved using `rENM_project_dir()`.

Details

This function is part of the rENM.model framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Inputs: A four-letter species code is required. The project directory is resolved either from the provided argument or via `rENM_project_dir()`.

Processing steps:

- Validates `alpha_code` and prepares the run directory
- Resolves the base project directory
- Spawns a PSOCK cluster with size: `min(number of years, detectCores())`
- Ensures each worker can call `create_ensemble_model()` within the rENM.model namespace
- Launches one job per 5-year bin using `parallel::parLapplyLB()` for load-balanced execution
- Collates per-year results including status, elapsed time, and output summaries
- Appends a formatted summary to the run log

Worker function availability: Workers in a PSOCK cluster are fresh R sessions. The function `create_ensemble_model()` is explicitly referenced from the rENM.model namespace and made available to workers via closure.

This supports both development (devtools::load_all()) and installed package workflows.

Fault tolerance: Each year is processed independently. Errors for individual years are captured and logged without aborting the full time series. The cluster is always stopped on exit.

Outputs: A processing summary is appended to:

`file.path(project_dir, "runs", <alpha_code>, "_log.txt")`

Value

Invisibly returns a named List keyed by year. Each element is:

- ok: Logical indicating successful completion
- year: Integer representing the 5-year bin
- elapsed: Numeric elapsed seconds for the task
- result: Output from `create_ensemble_model()` or NULL
- notes: Character containing "ok" or an error message

Attributes:

- total_elapsed_sec: Numeric total wall-clock time in seconds

Side effects:

- Executes parallel jobs using a PSOCK cluster
- Appends a processing summary to the run log
- Writes progress messages to the console

See Also

[create_ensemble_model()], **parallel**

Examples

```
## Not run:
  create_timeseries("CASP")
  create_timeseries("CASP", project_dir = "/my/project")

## End(Not run)
```

`plot_suitability` *Plot a climatic suitability raster with state boundaries and species range*

Description

Build a ggplot2 map of continuous climatic suitability values from a raster and overlay state or administrative boundaries plus a species range polygon. If the raster has multiple layers, layer selects which to plot.

Usage

```
plot_suitability(
  en,
  alpha_code,
  state_shp = "~/rENM/data/shapefiles/tl_2012_us_state/tl_2012_us_state.shp",
  title = "Climatic Suitability Map",
  limits = c(0, 1),
  layer = 1L,
  minor_div = 2L
)
```

Arguments

`en`	SpatRaster, Character, Raster. A suitability raster. One of: a terra SpatRaster, a single-file path (character) to a raster readable by terra rast(), or a raster::Raster* object converted internally to SpatRaster.
`alpha_code`	Character. Species code used to look up GAP.RANGE via get_species_info(alpha_code) and, from that, the species range shapefile under <project_dir>/data/shapefiles/<gap_range>/ <gap_range>.shp
`state_shp`	Character. Path to a states or administrative boundaries shapefile (.shp). Defaults to <project_dir>/data/shapefiles/tl_2012_us_state/ tl_2012_us_state.shp
`title`	Character. Plot title. Default "Climatic Suitability Map".
`limits`	Numeric. Length-2 vector giving the lower and upper limits of the color scale (default c(0, 1)).
`layer`	Integer. Band index to plot when en has multiple layers (default 1L).
`minor_div`	Integer. Number of minor grid divisions between major axis breaks (default 2L).

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context Build a ggplot2 map of continuous climatic suitability values from a raster (e.g., 0-1) and overlay state or administrative boundaries plus a species range polygon. If the raster has multiple layers, `layer` selects which to plot.

The species range shapefile is looked up via an internal helper `get_species_info(alpha_code)`, which must return a component `GAP.RANGE`. The shapefile is then read from: `<project_dir>/data/shapefiles/<gap_` `<gap_range>.shp`

Inputs

- A suitability raster provided as a SpatRaster, file path, or raster object.
- Species code used to look up GAP.RANGE via `get_species_info(alpha_code)`.
- Path to a states or administrative boundaries shapefile.

Outputs

- A ggplot object representing the climatic suitability surface with state boundaries and species range overlays.

Methods

- The state and species-range vectors are reprojected to the raster CRS (if set) to ensure alignment.
- The color scale uses a smooth multi-hue palette via `grDevices::colorRampPalette()`.
- NA cell values render transparent.

Data requirements

- Species range shapefile located at: `<project_dir>/data/shapefiles/<gap_range>/` `<gap_range>.shp`
- Default state shapefile located at: `<project_dir>/data/shapefiles/tl_2012_us_state/` `tl_2012_us_state.shp`

Value

A ggplot object. Side effects:

- None. The function returns a plot object without writing files.

Examples

```
## Not run:
# Example usage:
plot_suitability(en, "CASP")

## End(Not run)
```

reduce_covariance *Reduce variable covariance*

Description

Detects multicollinearity in the stagged predictor rasters through an adaptive Variable Inflation Factor (VIF) routine, records results, and retains only non-correlated variables for further processing.

Usage

```
reduce_covariance(
  alpha_code,
  year = NULL,
  vif_threshold = 10,
  sample_schedule = c(10000, 20000, 50000),
  stability_patience = 1,
  corr_threshold = 0.95,
  exclude_vars = character(0),
  file_patterns = c("\\.tif$", "\\.grd$", "\\.img$", "\\.bil$"),
  overwrite_results = TRUE
)
```

Arguments

`alpha_code`	Character. Species alpha code (for example, `"CASP"`).
`year`	Integer vector, or `NULL`. Years to process; if `NULL` (default) the sequence 1980, 1985, ..., 2020 is used.
`vif_threshold`	Numeric. VIF threshold passed to `usdm::vifstep()` (default 10).
`sample_schedule`	Integer vector. Adaptive sample sizes tried in order; the routine stops early once the kept set stabilizes. Default `c(10000, 20000, 50000)`.
`stability_patience`	Integer. Consecutive identical kept sets needed for convergence (default 1).
`corr_threshold`	Numeric. Absolute Pearson correlation threshold used when listing high-correlation pairs (default 0.95).
`exclude_vars`	Character vector. Predictor names that must always be retained. Case-sensitive match against sampled column names (default `character(0)`).
`file_patterns`	Character vector. Regular expressions selecting raster filenames to include. Matching stems trigger companion `.asc` moves. Default `c("\\.tif$", "\\.grd$", "\\.img$", "\\.bil$")`.
`overwrite_results`	Logical. Overwrite an existing results CSV (default `TRUE`).

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context

- Input rasters: `<project_dir>/runs/<alpha_code>/TimeSeries/ <year>/vars`
- Log file: `<project_dir>/runs/<alpha_code>/_log.txt`

Inputs

- Predictor rasters matching `file_patterns`
- Optional exclusions via `exclude_vars`

Outputs

- Results CSV: `_<alpha_code>-<year>-Correlation-Results.csv`
- Directory of moved rasters (only if variables are dropped): `_<alpha_code>-<year>-Correlated-Variables`
- Log entry appended to `_log.txt`

Methods

- Adaptive sampling over `sample_schedule` sizes until the kept set is stable for `stability_patience` iterations
- Multicollinearity screening with `usdm::vifstep()` at threshold `vif_threshold`
- Reporting of high-correlation pairs where $|r| \geq corr_threshold$

Data requirements

- Raster layers must share identical geometry
- Layer names must correspond to predictor identifiers

Value

Invisible list keyed by year containing:

- `kept` – character vector of retained predictors
- `dropped` – character vector of removed predictors
- `vif_table` – data frame of VIF results for all variables
- `cor_pairs` – data frame of high-correlation pairs
- `results_csv` – file path to the written CSV
- `moved_to` – directory path for relocated rasters or `NA` when none were moved

Examples

```
## Not run:
  reduce_covariance("CASP")
  reduce_covariance("CASP", year = 2005,
                    exclude_vars = c("bio1", "bio12"))
  reduce_covariance("CASP", year = 2000,
                    sample_schedule = c(8000, 16000, 32000, 64000))

## End(Not run)
```

save_suitability_plot *Save a ggplot suitability plot to file*

Description

Save a ggplot object to disk with configurable size, resolution, and typography. File format is inferred from the extension or defaults to PNG.

Usage

```
save_suitability_plot(
  p,
  fn,
  width = 1000,
  height = 750,
  res = 72,
  pointsize = 15,
  title_size = 28,
  subtitle_size = 20,
  legend_size = 15,
  legend_title_size = 15,
  axis_title_size = 20,
  axis_text_size = 18
)
```

Arguments

p	ggplot. A ggplot object to save.
fn	Character. Full output path and file name, with or without an extension. If no extension is provided, ".png" is appended.
width	Numeric. Image width in pixels. Default 1000.
height	Numeric. Image height in pixels. Default 750.
res	Numeric. Resolution (DPI). Default 72.
pointsize	Numeric. Base point size for text passed to ggsave(). Default 15.
title_size	Numeric. Font size for the plot title. Default 28.
subtitle_size	Numeric. Font size for the plot subtitle. Default 20.
legend_size	Numeric. Font size for legend text. Default 15.
legend_title_size	Numeric. Font size for the legend title. Default 15.
axis_title_size	Numeric. Font size for axis titles. Default 20.
axis_text_size	Numeric. Font size for axis tick labels. Default 18.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context This function saves ggplot outputs generated during the rENM workflow, resolving relative paths against the project directory defined by rENM_project_dir().

Inputs

- A ggplot object.
- An output file path, with or without extension.

Outputs

- A saved image file written to disk.
- If no extension is provided, a PNG file is written.

Methods Width and height are specified in pixels and converted to inches using the resolution (DPI) for ggplot2::ggsave().

Supported extensions:

- "png", "pdf", "tiff", "jpg", "jpeg", "svg", "eps"

Before saving, the plot theme is enhanced to apply specified font sizes to the title, subtitle, legend, and axis text elements.

Data requirements The output path is resolved relative to the project directory if a non-absolute path is provided.

Value

Invisibly returns the file path written.

- Character. File path to the saved plot.
- Side effect: Writes an image file to disk.

See Also

[ggplot2::ggsave()], [ggplot2::theme()], [ggplot2::element_text()]

Examples

```
## Not run:
  p <- ggplot2::ggplot(mtcars, ggplot2::aes(mpg, wt)) +
         ggplot2::geom_point()
  save_suitability_plot(p, "plots/mtcars_scatter.png")

  # No extension provided -> defaults to PNG:
  save_suitability_plot(p, "plots/mtcars_scatter")

## End(Not run)
```

screen_by_convergence1 — *Iterative bivariate ENM screening with adaptive, stability-aware convergence*

Description

Repeatedly fits bivariate MaxEnt models, aggregates permutation importance (PI) scores, and stops when both the mean PI and membership of the adaptive top-k variable set stabilize. Designed for rapid, reproducible predictor down-selection across 5-year intervals.

Usage

```
screen_by_convergence1(
  alpha_code,
  year = NULL,
  max_iter = 50,
  batch_samples = NULL,
  w_stability = 0.5,
  background_n = 2500,
  seed = NULL,
  ncores = max(1, parallel::detectCores() - 1),
  fast = FALSE,
  set_threshold = 0.9,
  passes_required = NULL
)
```

Arguments

`alpha_code`	Character. Species or run code (e.g., "CASP").
`year`	Integer, Character, or NULL. Year to process; NULL runs all 5-year intervals 1980–2020.
`max_iter`	Integer. Maximum iterations safeguard; default 50.
`batch_samples`	Integer or NULL. Target appearances per variable per iteration; NULL triggers an adaptive fallback.
`w_stability`	Numeric. Instability weight in `wz_score`; default 0.5.
`background_n`	Integer. Background points per model; default 2500.
`seed`	Integer or NULL. RNG seed; NULL draws a random seed that is returned.
`ncores`	Integer. CPU cores for parallel fits; default `max(1, parallel::detectCores() - 1)`.
`fast`	Logical. Enable fast triage mode; default FALSE.
`set_threshold`	Numeric. Jaccard threshold for membership stability; default 0.90. Must lie in `(0, 1)`.
`passes_required`	Integer or NULL. Consecutive passes needed for convergence; NULL defaults to 1 in fast mode, else 2.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context

- Reads occurrences from `runs/<alpha_code>/_occs/of-<year>.csv`
- Reads raster predictors from `runs/<alpha_code>/_vars/<year>/`
- Writes outputs to `runs/<alpha_code>/TimeSeries/<year>/vars/`

Inputs

- Occurrence table with `longitude`, `latitude`
- At least two GeoTIFF predictor layers

Methods Convergence requires both tests to pass:

- *Mean stability* – relative change in top-k mean PI below an adaptive threshold (`10%` if $p \leq 20$, `7.5%` if $21 \leq p \leq 40$, else `5%`).
- *Membership stability* – Jaccard overlap of consecutive top-k sets above `set_threshold`.

Adaptive top-k: $k = \min(12, \max(5, \mathrm{round}(\sqrt{p})))$. Batch sampling defaults to `max(1, floor(1.5 * sqrt(p)))` appearances per variable and is halved in `fast` mode. Parallel execution uses deterministic RNG streams.

Outputs

- Variable ranking CSV
- Convergence trace CSV
- Log entry appended to `_log.txt`

If `year = NULL`, all 5-year intervals 1980–2020 are processed.

Value

A list (invisible) with:

`importance_summary` Data frame of variable-wise PI statistics and rank diagnostics.

`convergence_trace` Data frame with per-iteration diagnostics.

`output_files` Named vector with paths to the two CSV files.

`seed_used` Integer seed actually used in this run.

Side effects: two CSV files and a log block are written to disk inside the project directory.

See Also

`maxent`, `stack`

Examples

```
## Not run:
## Quick triage across all years
res <- screen_by_convergence1("CASP", fast = TRUE)

## Standard fidelity for a single year
res <- screen_by_convergence1("CASP", 1990, seed = 42)

## End(Not run)
```

---

screen_by_convergence2 — *Iterative bivariate ENM screening with adaptive, stability-aware convergence*

Description

Repeatedly fits bivariate maxnet ecological niche models, aggregates permutation importance (PI) scores, and stops when both the mean PI and membership of the adaptive top-k variable set stabilize. Designed for rapid, reproducible predictor down-selection across 5-year intervals without the Java dependency required by dismo::maxent().

Usage

```
screen_by_convergence2(
  alpha_code,
  year = NULL,
  max_iter = 50,
  batch_samples = NULL,
  w_stability = 0.5,
  background_n = 2500,
  seed = NULL,
  ncores = max(1, parallel::detectCores() - 1),
  fast = FALSE,
  set_threshold = 0.9,
  passes_required = NULL
)
```

Arguments

alpha_code	Character. Species or run code (e.g., "CASP").
year	Integer, Character, or NULL. Year to process; NULL runs all 5-year intervals 1980-2020.
max_iter	Integer. Maximum iterations safeguard; default 50.
batch_samples	Integer or NULL. Target appearances per variable per iteration; NULL triggers an adaptive fallback.
w_stability	Numeric. Instability weight in wz_score; default 0.5.
background_n	Integer. Background points per model; default 2500.

`seed` Integer or NULL. RNG seed; NULL draws a random seed that is returned.

`ncores` Integer. CPU cores for parallel fits; default `max(1, parallel::detectCores() - 1)`.

`fast` Logical. Enable fast triage mode; default FALSE.

`set_threshold` Numeric. Jaccard threshold for membership stability; default 0.90. Must lie in `(0, 1)`.

`passes_required` Integer or NULL. Consecutive passes needed for convergence; NULL defaults to 1 in fast mode, else 2.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by `rENM_project_dir()`.

Pipeline context

- Reads occurrences from runs/<alpha_code>/_occs/of-<year>.csv
- Reads raster predictors from runs/<alpha_code>/_vars/<year>/
- Writes outputs to runs/<alpha_code>/TimeSeries/<year>/vars/

Inputs

- Occurrence table with `longitude`, `latitude`
- At least two GeoTIFF predictor layers

Methods The function repeatedly samples bivariate predictor pairs and fits `maxnet` models using occurrence and background environmental values extracted from the selected raster layers. Background points are drawn directly from non-`NA` raster cells using `raster::sampleRandom(xy = TRUE, na.rm = TRUE)`, eliminating the need for `dismo::randomPoints()`.

For each fitted pair, permutation importance is computed manually in a MaxEnt-like way: each predictor is permuted in turn on the model evaluation table, predictions are recomputed without refitting, the drop in training AUC is measured, and the resulting drops are normalized to percentages summing to 100 across the two predictors in that bivariate model. This produces a permutation-based importance metric analogous in spirit to Java MaxEnt's permutation importance, while remaining fully native to R.

Convergence requires both tests to pass:

- *Mean stability* - relative change in top-k mean PI below an adaptive threshold (`10%` if $p \leq 20$, `7.5%` if $21 \leq p \leq 40$, else `5%`).
- *Membership stability* - Jaccard overlap of consecutive top-k sets above `set_threshold`.

Adaptive top-k: $k = \min(12, \max(5, \mathrm{round}(\sqrt{p})))$. Batch sampling defaults to `max(1, floor(1.5 * sqrt(p)))` appearances per variable and is halved in `fast` mode. Parallel execution uses deterministic RNG streams.

Outputs

- Variable ranking CSV
- Convergence trace CSV
- Log entry appended to `_log.txt`

If `year = NULL`, all 5-year intervals 1980-2020 are processed.

Value

A list (invisible) with:

`importance_summary` Data frame of variable-wise PI statistics and rank diagnostics.

`convergence_trace` Data frame with per-iteration diagnostics.

`output_files` Named vector with paths to the two CSV files.

`seed_used` Integer seed actually used in this run.

Side effects: two CSV files and a log block are written to disk inside the project directory.

See Also

`maxnet`, `stack`

Examples

```
## Not run:
## Quick triage across all years
res <- screen_by_convergence2("CASP", fast = TRUE)

## Standard fidelity for a single year
res <- screen_by_convergence2("CASP", 1990, seed = 42)

## End(Not run)
```

`stage_all_variables` *Stage environmental variables into TimeSeries bins*

Description

Copies raster variables from run-level storage into TimeSeries bins, organizing predictors for downstream modeling while reporting progress and recording a processing summary in the run log.

Usage

```
stage_all_variables(
  alpha_code,
  year = seq(1980, 2020, by = 5),
  project_dir = NULL
)
```

Arguments

`alpha_code`	Character. Species alpha code (e.g., "CASP").
`year`	Integer. Vector of 5-year bins. Default: `seq(1980, 2020, by = 5)`.
`project_dir`	Character, NULL. Optional project root directory. If NULL, resolved using `rENM_project_dir()`.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Inputs: Copies raster variables ('.tif', '.asc') from the run-level '_vars' directory into the corresponding TimeSeries directories:

<project_dir>/runs/<alpha_code>/ _vars/<year>/*.(tif|asc)

Outputs: Files are staged into:

<project_dir>/runs/<alpha_code>/ TimeSeries/<year>/vars/*

Existing files are overwritten.

Methods:

- Recursively scans source directories for '.tif' and '.asc' files
- Copies files into the target directory (flat structure)
- Overwrites existing files
- Warns on missing source directories
- Warns on duplicate basenames (last file wins)

Logging: A processing summary is appended to:

<project_dir>/runs/<alpha_code>/ _log.txt

including:

- Timestamp
- Alpha code
- Number of files transferred
- File list
- Total elapsed time
- Output location

Value

Data frame (invisibly returned) with columns:

- `year`
- `src_dir`
- `dest_dir`
- `files_found`
- `files_copied`
- `duplicates`

Side effects:

- Copies raster files into TimeSeries directories
- Overwrites existing files in target directories
- Appends a processing summary to the run log
- Writes progress messages to the console

Examples

```
## Not run:
  stage_all_variables("CASP")
  stage_all_variables("CASP", year = 2005)
  stage_all_variables("CASP", year = c(1990, 2010, 2020))
  stage_all_variables("CASP", project_dir = "/projects/rENM")

## End(Not run)
```

---

stage_occurrences *Stage occurrence CSVs into time series bins*

Description

Copies per-bin occurrence files into the TimeSeries structure for downstream modeling while ensuring consistency with preprocessing outputs and recording a detailed audit log.

Usage

```
stage_occurrences(alpha_code, year = seq(1980, 2020, 5), project_dir = NULL)
```

Arguments

`alpha_code`	Character. Species alpha code (e.g., "CASP").
`year`	Integer. Vector of 5-year bins. Default: `seq(1980, 2020, 5)`.
`project_dir`	Character, NULL. Optional project root directory. If NULL, resolved using `rENM_project_dir()`.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Inputs: Copies per-bin occurrence files (`of-<year>.csv`) from:

`<project_dir>/runs/<alpha_code>/ _occs/of-<year>.csv`

Outputs: Files are staged into:

`<project_dir>/runs/<alpha_code>/ TimeSeries/<year>/occs/of-<year>.csv`

Existing files are overwritten to ensure consistency with the latest preprocessing outputs.

Methods:

- Validates requested years (must be (1980, 2020), divisible by 5)
- Creates destination directories as needed
- Copies files using `file.copy(..., overwrite = TRUE)`
- Records per-year status: Copied, Missing, or Error
- Aggregates results into a structured log block

Logging: A standardized summary is appended to:

`<project_dir>/runs/<alpha_code>/ _log.txt`

including:

- Timestamp and alpha code
- Per-year staging table
- Counts (total, copied, missing)
- Output paths (multi-line)
- Total elapsed time

Each run is separated by a 72-character divider for readability.

Value

List (invisibly returned) with elements:

- `alpha_code`: Character species alpha code
- `years`: Integer vector of processed years
- `copied_paths`: Character vector of copied file paths
- `counts`: List with elements total_years, valid_files, copied_files, missing_files
- `elapsed_seconds`: Numeric elapsed time in seconds
- `log_file`: Character path to the run log file

Side effects:

- Copies occurrence CSV files into TimeSeries directories
- Overwrites existing files in destination directories
- Appends a processing summary to the run log
- Writes audit output to the console

Examples

```
## Not run:
  stage_occurrences("CASP")
  stage_occurrences("CASP", year = 1990)
  stage_occurrences("CASP", year = c(1980, 2000, 2015))

## End(Not run)
```

`stage_screened_variables` *Stage screened environmental variables into the time series*

Description

Stage (copy) ranked raster predictor layers into per-year TimeSeries directories based on a master variable-ranking table. Supports several ranking metrics and an optional top-N filter.

Usage

```
stage_screened_variables(
  alpha_code,
  year = seq(1980, 2020, 5),
  top_n = 10,
  rank_method = "ratio"
)
```

Arguments

alpha_code	Character. Species alpha code (e.g., "CASP"). Used to resolve paths under the project runs directory.
year	Integer. Vector of years or single year. Defaults to seq(1980, 2020, 5). Each value must be a 5-year step within 1980-2020.
top_n	Integer or NULL. Number of top-ranked variables to stage per year. Default = 10; use NULL to copy all variables.
rank_method	Character. One of "ranksum","medianPI","ratio","wz". Selects the ranking column used to order variables. Default = "ratio".

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Inputs

- Master ranking CSV, one per year: `<proj_root>/runs/<alpha_code>/TimeSeries/<year>/vars/_<alpha_code>-<year>-Variable-Ranking.csv`
- Source rasters: `<proj_root>/runs/<alpha_code>/_vars/<year>/`

Outputs

- Ranked rasters copied to: `<proj_root>/runs/<alpha_code>/TimeSeries/<year>/vars/`
- Per-year status line and a 72-dash summary appended to: `<proj_root>/runs/<alpha_code>/_log.txt`

Methods

- Ranking columns: `rank_ranksum`, `rank_medianPI`, `rank_ratio`, `rank_wz`
- Tie-breakers use the associated score column (higher or lower depending on metric).

Data requirements

- Years must be multiples of five in the range {1980, 1985, ..., 2020}.
- Raster filenames must match the `var` column in the ranking CSV.

Value

Invisible named list with:

- `alpha_code` Character
- `years` Integer vector
- `rank_method` Character
- `top_n` Integer or NULL

- `copied_paths` Character vector
- `counts` List: `total_years`, `attempted_files`, `copied_files`, `missing_files`, `years_missing_ranking`
- `elapsed_seconds` Numeric
- `log_file` Character

Examples

```
## Not run:
stage_screened_variables("CASP", top_n = 5, rank_method = "medianPI")

## End(Not run)
```

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